VHDL Assignment 3

Gate-level circuits, synthesized designs and simulation evidence

Five sequential circuits | GHDL 5.0.1 | Logisim-evolution 5.0.0 | VHDL-2008

Verification method

Each circuit has two independently executed implementations: a VHDL design compiled, elaborated, simulated and synthesized with GHDL, and an individual-gate Logisim project loaded into the actual Logisim simulation engine. No built-in latch or flip-flop components are used in the Logisim designs. All five circuits use visible wires for inputs, outputs, clocks and cross-feedback. There are no Tunnel, built-in Latch or built-in Flip-Flop components. All five projects also pass the same checks in Logisim-evolution 5.0.0.

Results

Circuit	VHDL simulation	Synthesis	Logisim steps
RS Latch	PASS	PASS	6
D Latch with Enable	PASS	PASS	6
Positive-edge D Flip-Flop	PASS	PASS	8
Positive-edge JK Flip-Flop	PASS	PASS	28
D Flip-Flop with Asynchronous Set/Clear	PASS	PASS	17

64 expected-Q checks pass in each implementation; one initial preparation step has no specified power-on Q. Six additional Qn checks verify the RS latch. The tests cover forbidden RS inputs, latch transparency/hold, rising and falling edges, all JK modes, asynchronous controls with both clock levels, priority and release behavior.

Scope and timing assumptions

These are ideal digital functional simulations, not transistor-level timing proofs. Uninitialized storage has no guaranteed power-on value. The revised JK circuit starts with its explicit clr_n input at 0, so this circuit is held in reset until clr_n is set to 1. Input values are held stable before sampling edges. Real hardware must satisfy setup/hold and asynchronous recovery/removal timing; asynchronous inputs must not be released near a clock edge. The RS forbidden-state release is deliberately not assigned a deterministic result. Feedback latches and dual asynchronous controls may need device-specific primitives for a particular FPGA target; synthesis here is generic GHDL synthesis, without FPGA place-and-route.

Submitted materials

The repository contains vhdl/ (five VHDL sources), logisim/ (five gate-level projects), this PDF report, and README.md. Verification used assertion-based testbenches, GHDL VCD traces, synthesized netlists and actual Logisim engine runs; the circuit diagrams, waveforms, execution excerpts and check tables are embedded in this report. Generated synthesis diagrams visualize the emitted netlist; they are not FPGA technology-mapped schematics.

1. RS Latch

Gate-level implementation

Cross-coupled NOR gates implement Q = NOT(R OR Qn), Qn = NOT(S OR Q). S,R = 10 sets; 01 resets; 00 holds. The forbidden 11 condition forces Q=Qn=0, so the outputs are not complementary. Simultaneous release of 11 to 00 has no defined physical result; the test recovers by applying a valid reset. On cold start with S=R=0, Logisim-evolution 5.0.0 displays E for the uninitialized feedback loop. Apply S=0, R=1 first, then R=0 to obtain a defined Hold state. The crossing without a dot is NOT a junction.

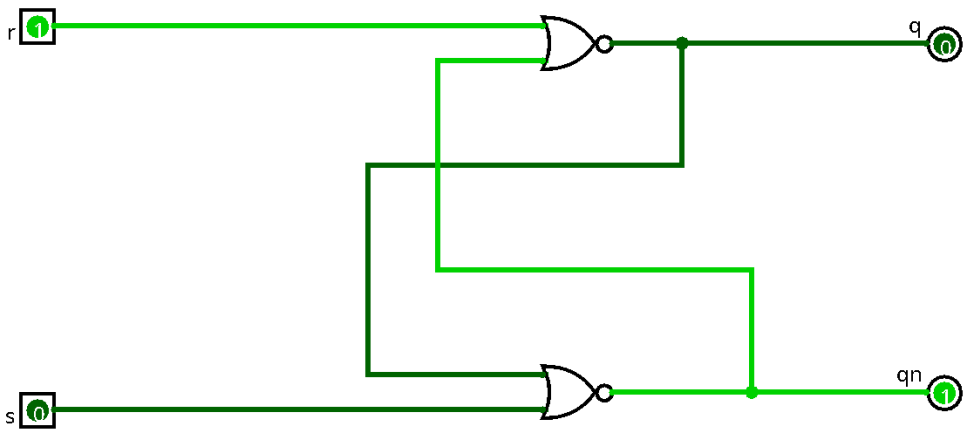


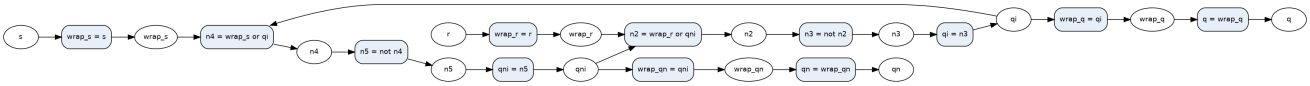
Figure: rendered directly from the submitted .circ project using Logisim. RS feedback is drawn with visible wires; the crossing without a dot is not connected. State shown: final valid Reset (Q=0, Qn=1).

Simulation checks

Step	Input case	Q	Expected	Result
1	Reset	0	0	PASS
2	Hold reset	0	0	PASS
3	Set	1	1	PASS
4	Hold set	1	1	PASS
5	Forbidden both outputs zero	0	0	PASS
6	Recover by reset	0	0	PASS

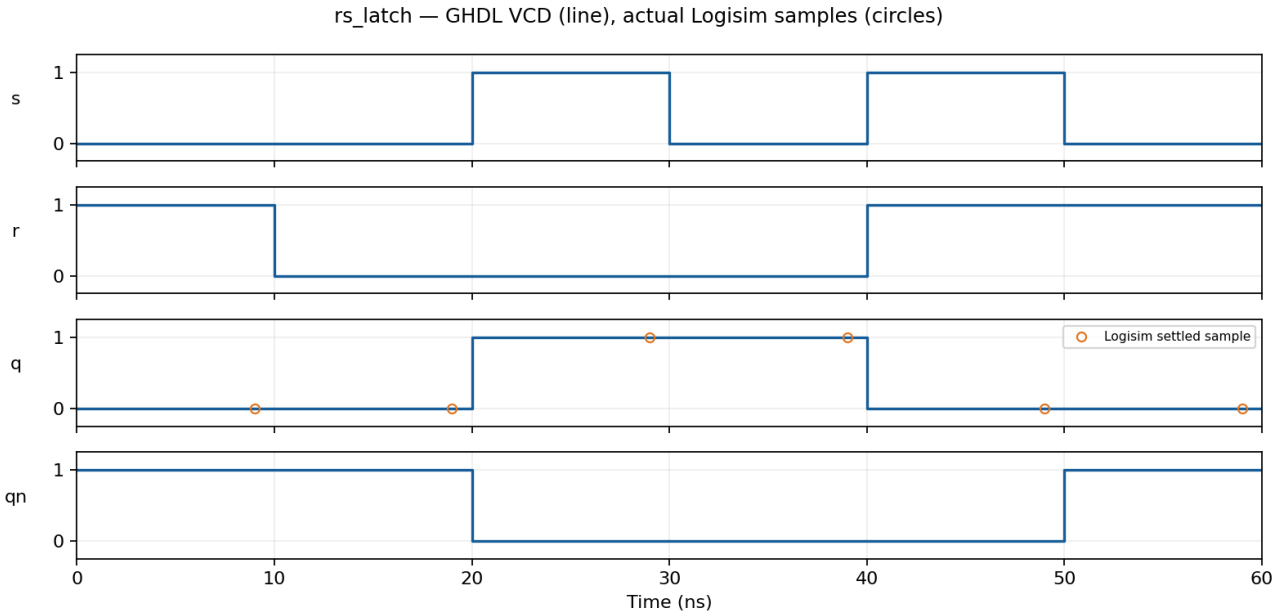
1. RS Latch — VHDL

Generated synthesis schematic



Generated from the GHDL generic synthesis netlist; boxes represent emitted concurrent assignments or sequential processes.

Simulation waveform



Blue: GHDL VCD traces. Orange circles: independently simulated Logisim Q after each input step (placed 1 ns before the corresponding VHDL step boundary for readability). Unknown initial values are marked U/X or shown at midlevel. Logisim time is propagation settling, so the circles compare logical states, not propagation delay.

VHDL verification

/workspace/output/assignment3/tb/tb_rs_latch.vhd:47:3:@60ns:(report note): PASS rs_latch
simulation stopped @60ns

All RTL and synthesized-netlist assertions passed; generic synthesis completed successfully. Source: vhdl/rs_latch.vhd. Assertion-based testbench results are shown above.

2. D Latch with Enable

Gate-level implementation

A NAND inverter creates Dn. Sbar=NAND(D,en) and Rbar=NAND(Dn,en) drive a cross-coupled NAND latch. When en=1, Q follows D; when en=0, both latch inputs become 1 and the previous state is held.

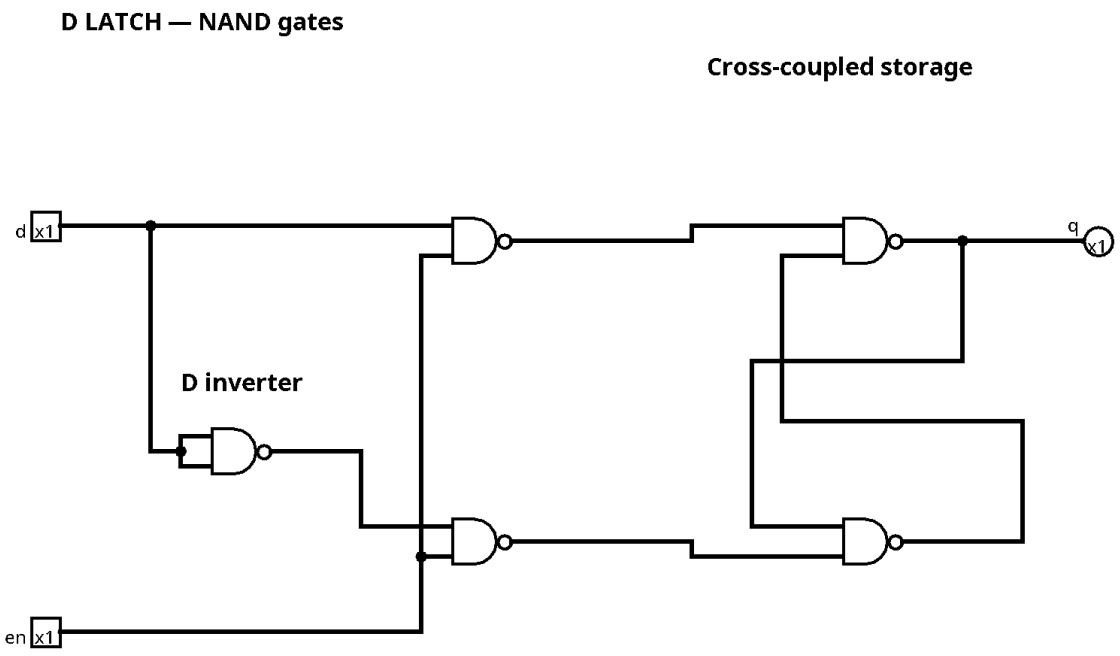


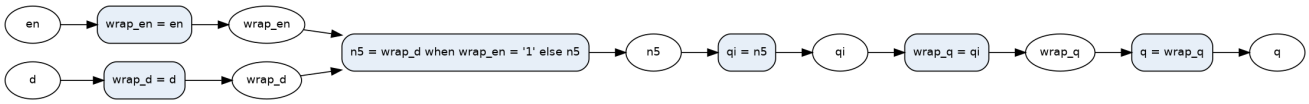
Figure: rendered directly from the submitted .circ project using Logisim. All connections and storage feedback are drawn as visible wires. A crossing without a dot is not a connection. The displayed states are hidden to emphasize connectivity.

Simulation checks

Step	Input case	Q	Expected	Result
1	Initialize	0	0	PASS
2	Transparent high	1	1	PASS
3	Transparent low	0	0	PASS
4	Close	0	0	PASS
5	Hold despite D change	0	0	PASS
6	Reopen	1	1	PASS

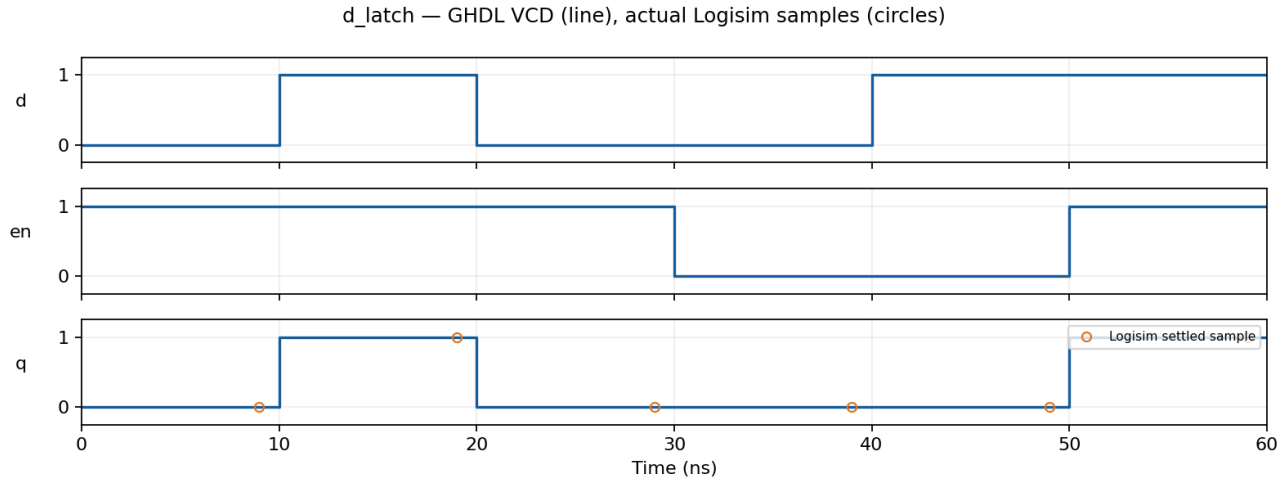
2. D Latch with Enable — VHDL

Generated synthesis schematic



Generated from the GHDL generic synthesis netlist; boxes represent emitted concurrent assignments or sequential processes.

Simulation waveform



Blue: GHDL VCD traces. Orange circles: independently simulated Logisim Q after each input step (placed 1 ns before the corresponding VHDL step boundary for readability). Unknown initial values are marked U/X or shown at midlevel. Logisim time is propagation settling, so the circles compare logical states, not propagation delay.

VHDL verification

/workspace/output/assignment3/tb/tb_d_latch.vhd:38:3:@60ns:(report note): PASS d_latch
simulation stopped @60ns

All RTL and synthesized-netlist assertions passed; generic synthesis completed successfully. Source: vhdl/d_latch.vhd. Assertion-based testbench results are shown above.

3. Positive-edge D Flip-Flop

Gate-level implementation

Two gate-built D latches form a master-slave flip-flop. The master is enabled by NOT(clk), and the slave by clk. The master samples during the low phase; the slave transfers that stored value on the rising edge. Changes to D during the high phase do not reach Q.

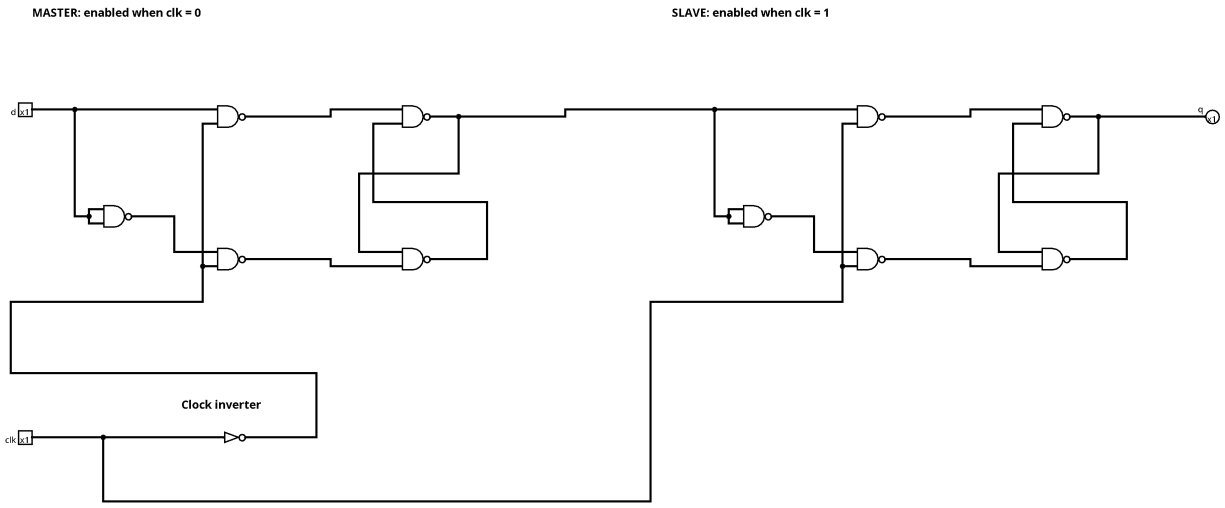


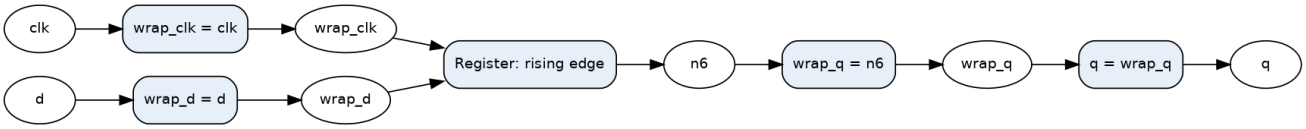
Figure: rendered directly from the submitted .circ project using Logisim. All connections and storage feedback are drawn as visible wires. A crossing without a dot is not a connection. The displayed states are hidden to emphasize connectivity.

Simulation checks

Step	Input case	Q	Expected	Result
1	Initialize low phase	E	-	PASS
2	Rising capture zero	0	0	PASS
3	High phase hold	0	0	PASS
4	Falling edge hold	0	0	PASS
5	Rising capture one	1	1	PASS
6	High phase hold one	1	1	PASS
7	Falling hold one	1	1	PASS
8	Rising capture zero	0	0	PASS

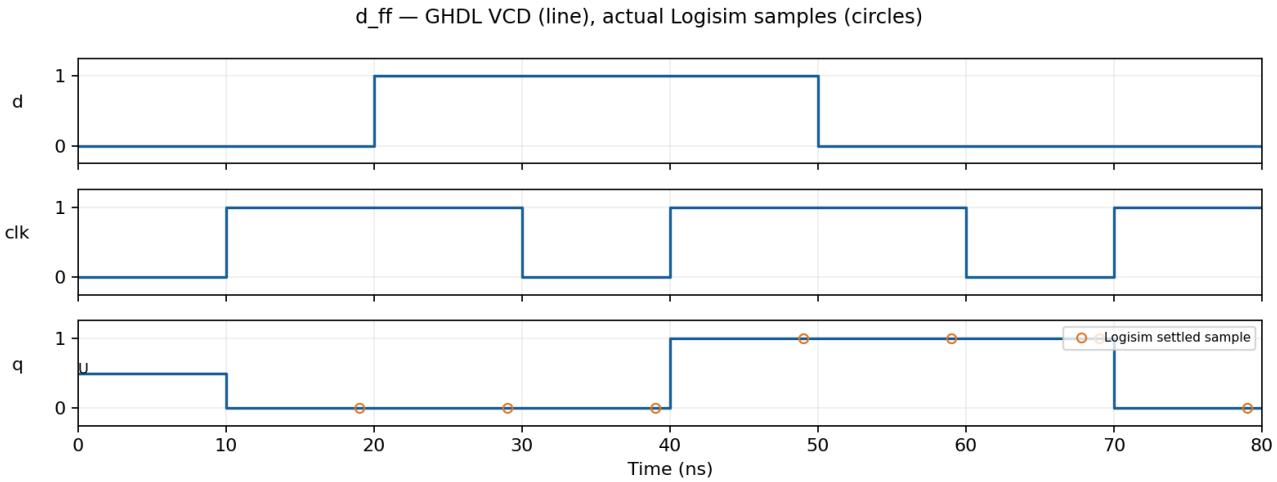
3. Positive-edge D Flip-Flop — VHDL

Generated synthesis schematic



Generated from the GHDL generic synthesis netlist; boxes represent emitted concurrent assignments or sequential processes.

Simulation waveform



Blue: GHDL VCD traces. Orange circles: independently simulated Logisim Q after each input step (placed 1 ns before the corresponding VHDL step boundary for readability). Unknown initial values are marked U/X or shown at midlevel. Logisim time is propagation settling, so the circles compare logical states, not propagation delay.

VHDL verification

/workspace/output/assignment3/tb/tb_d_ff.vhd:43:3:@80ns:(report note): PASS d_ff
simulation stopped @80ns

All RTL and synthesized-netlist assertions passed; generic synthesis completed successfully. Source: vhdl/d_ff.vhd. Assertion-based testbench results are shown above.

4. Positive-edge JK Flip-Flop

Gate-level implementation

The feedback next-state equation is Dnext=(J AND Qn) OR ((NOT K) AND Q). NAND gates implement this equation and two NAND D latches form the master-slave storage. JK=00 holds, 01 resets, 10 sets, and 11 toggles once per rising edge. An added active-low clr_n input initializes both latch stages. At startup clr_n=0 forces Q=0 and all 15 NAND outputs are defined. Set clr_n=1 to run. In each latch, clr_n is an additional input of both data-gating NANDs and the Qn storage NAND. Clear works with either clock level. Both VHDL and Logisim implement the same added asynchronous clear.

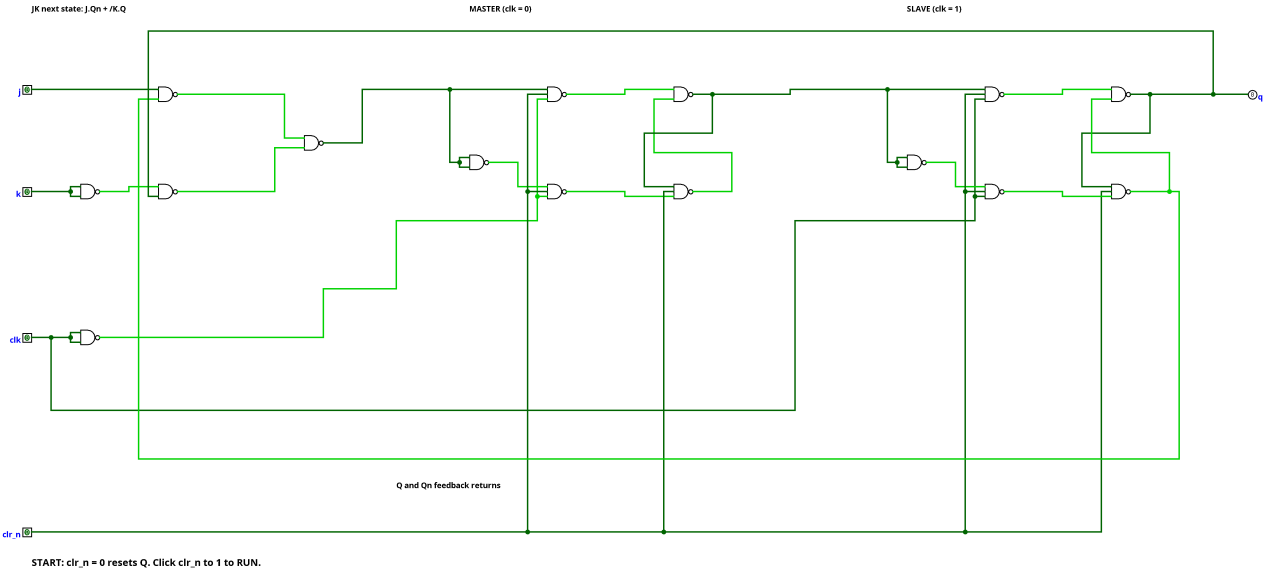


Figure: rendered directly from the submitted .circ project using Logisim. All connections and storage feedback are visible. Shown state: cold start with clr_n=0; all gate outputs defined, Q=0. Use clr_n=1 for normal JK operation.

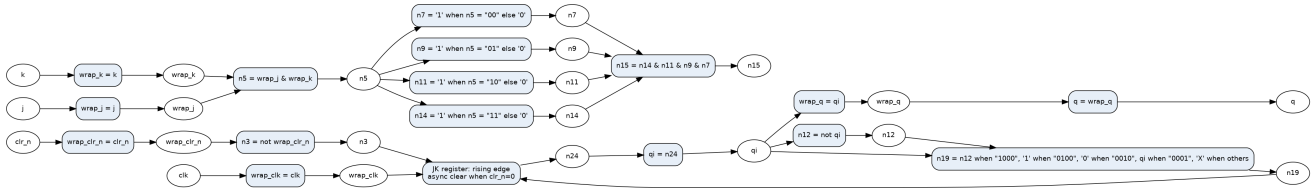
Simulation checks

Step	Input case	Q	Expected	Result
1	Power-on clear	0	0	PASS
2	Release clear low clock	0	0	PASS
3	Prepare Set	0	0	PASS
4	Set	1	1	PASS
5	Input change at high clock holds	1	1	PASS
6	Falling edge holds	1	1	PASS
7	Reset	0	0	PASS
8	Prepare Hold zero	0	0	PASS
9	Hold zero	0	0	PASS
10	Prepare Set again	0	0	PASS
11	Set again	1	1	PASS
12	Prepare Hold one	1	1	PASS
13	Hold one	1	1	PASS
14	Prepare Toggle	1	1	PASS
15	Toggle to zero	0	0	PASS
16	High clock stays constant	0	0	PASS
17	Falling edge holds zero	0	0	PASS

Step	Input case	Q	Expected	Result
18	Toggle to one	1	1	PASS
19	Asynchronous clear at high clock	0	0	PASS
20	Release clear high clock holds	0	0	PASS
21	Prepare Toggle after clear	0	0	PASS
22	Toggle after clear	1	1	PASS
23	Falling edge holds one	1	1	PASS
24	Asynchronous clear at low clock	0	0	PASS
25	Clear overrides Set and edge	0	0	PASS
26	Clear release without edge holds	0	0	PASS
27	Prepare normal Set	0	0	PASS
28	Normal Set restored	1	1	PASS

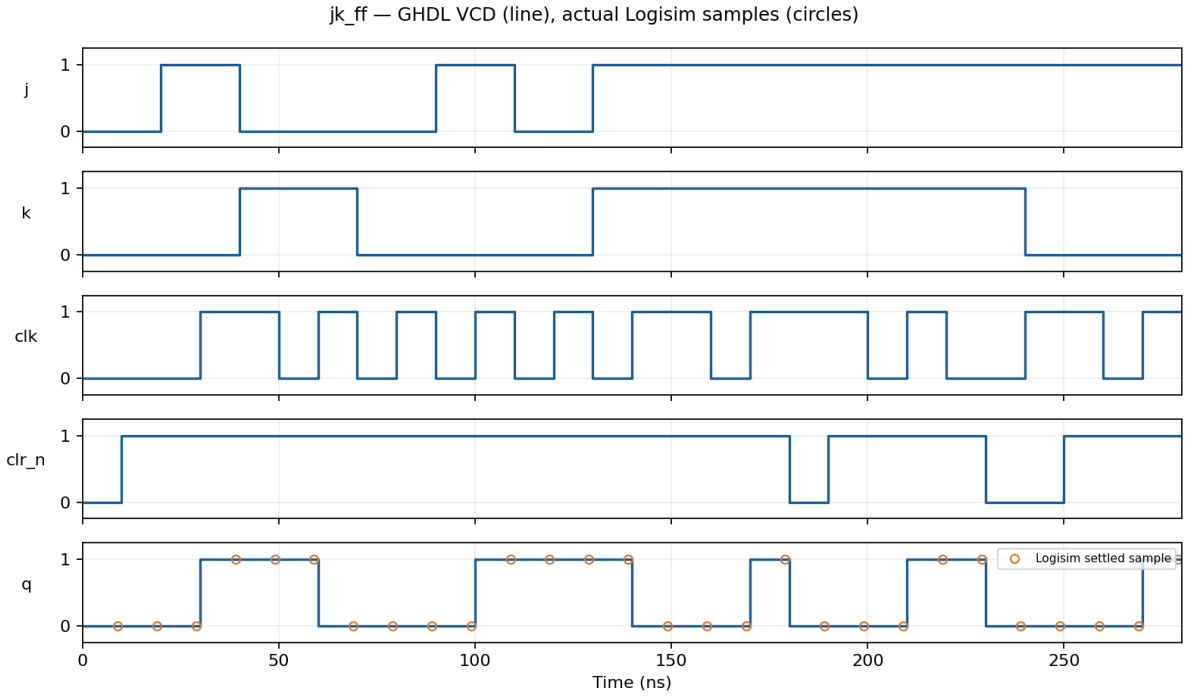
4. Positive-edge JK Flip-Flop — VHDL

Generated synthesis schematic



Generated from the GHDL generic synthesis netlist; boxes represent emitted concurrent assignments or sequential processes.

Simulation waveform



Blue: GHDL VCD traces. Orange circles: independently simulated Logisim Q after each input step (placed 1 ns before the corresponding VHDL step boundary for readability). Unknown initial values are marked U/X or shown at midlevel. Logisim time is propagation settling, so the circles compare logical states, not propagation delay.

VHDL verification

/workspace/output/assignment3/tb/tb_jk_ff.vhd:105:1:@280ns:(report note): PASS jk_ff with active-low clear: 28 checks simulation stopped @280ns

All RTL and synthesized-netlist assertions passed; generic synthesis completed successfully. Source: vhdl/jk_ff.vhd. Assertion-based testbench results are shown above.

5. D Flip-Flop with Asynchronous Set/Clear

Gate-level implementation

Both master and slave use gate-built NOR storage with asynchronous overrides. run=NOT(set OR clr), effective_set=set AND NOT(clr). For each latch, S=(D AND enable AND run) OR effective_set, R=(NOT(D) AND enable AND run) OR clr. Thus clr always wins. Applying controls to both stages prevents a stale master value from appearing when the controls are released during a high clock phase.

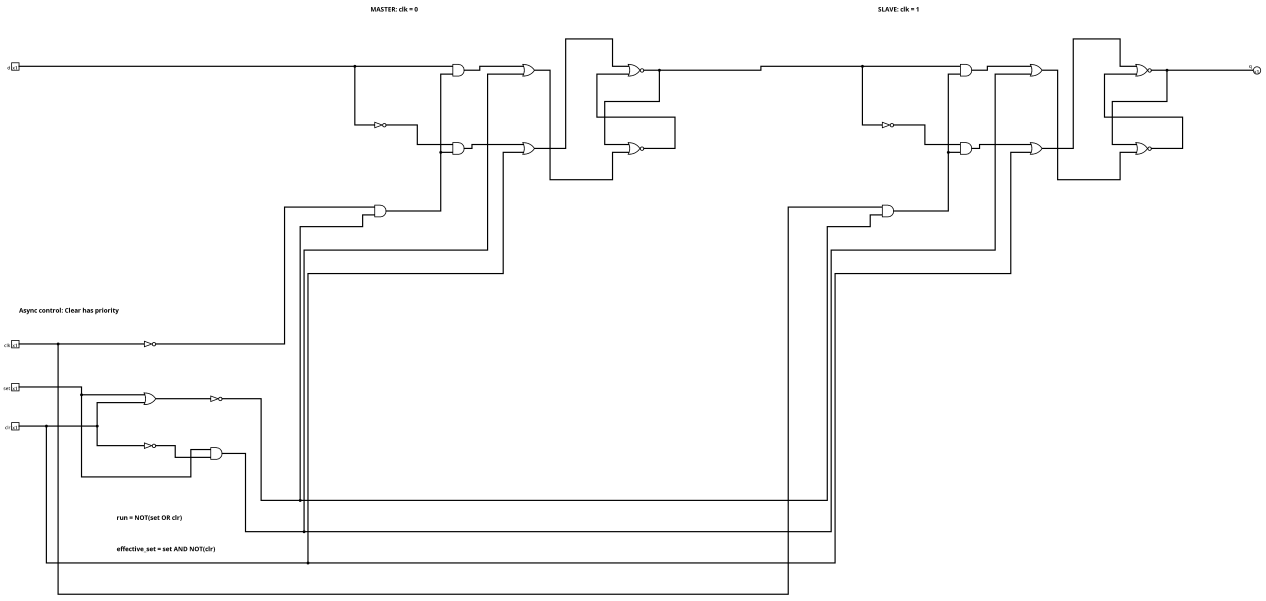


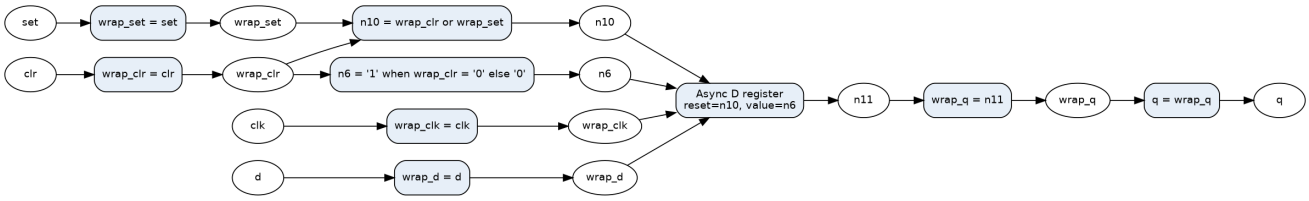
Figure: rendered directly from the submitted .circ project using Logisim. All connections and storage feedback are drawn as visible wires. A crossing without a dot is not a connection. The displayed states are hidden to emphasize connectivity.

Simulation checks

Step	Input case	Q	Expected	Result
1	Async clear low clock	0	0	PASS
2	Release clear	0	0	PASS
3	Async set low clock	1	1	PASS
4	Clear priority both active	0	0	PASS
5	Set remains asserted	1	1	PASS
6	Release set	1	1	PASS
7	Normal capture zero	0	0	PASS
8	High phase data hold	0	0	PASS
9	Async set high clock	1	1	PASS
10	Clear priority high clock	0	0	PASS
11	Release controls high clock	0	0	PASS
12	Falling hold	0	0	PASS
13	Normal capture one	1	1	PASS
14	Async clear high clock	0	0	PASS
15	Release clear high clock	0	0	PASS
16	Prepare zero	0	0	PASS
17	Capture zero again	0	0	PASS

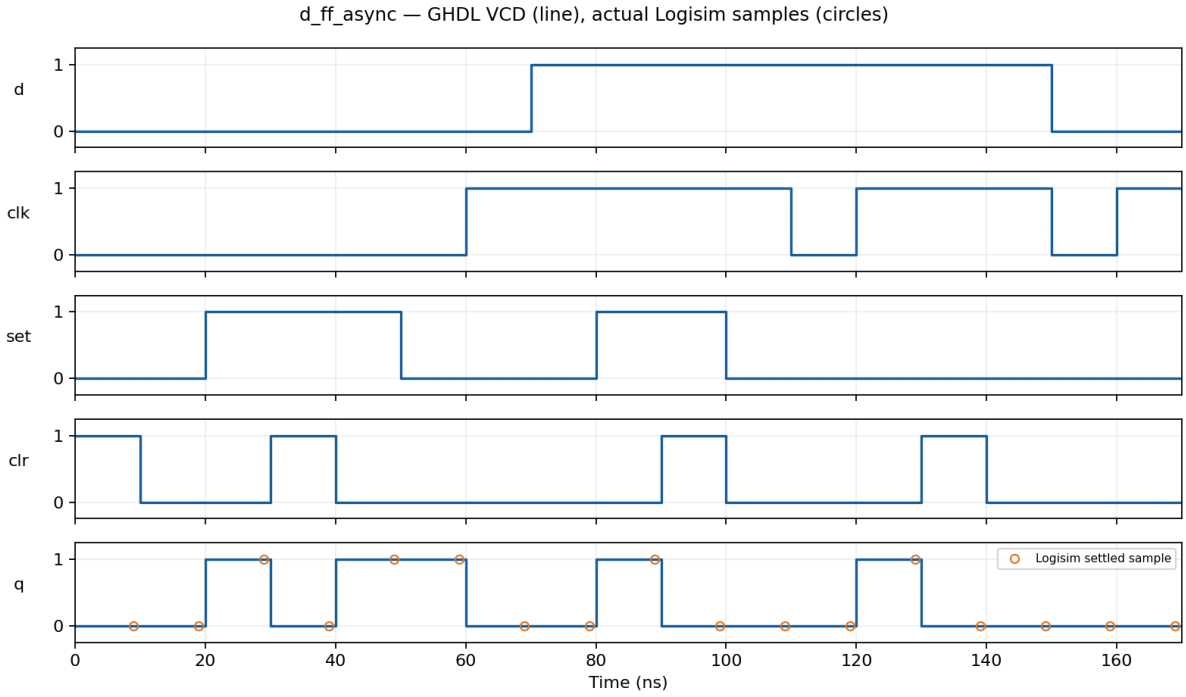
5. D Flip-Flop with Asynchronous Set/Clear — VHDL

Generated synthesis schematic



Generated from the GHDL generic synthesis netlist; boxes represent emitted concurrent assignments or sequential processes.

Simulation waveform



Blue: GHDL VCD traces. Orange circles: independently simulated Logisim Q after each input step (placed 1 ns before the corresponding VHDL step boundary for readability). Unknown initial values are marked U/X or shown at midlevel. Logisim time is propagation settling, so the circles compare logical states, not propagation delay.

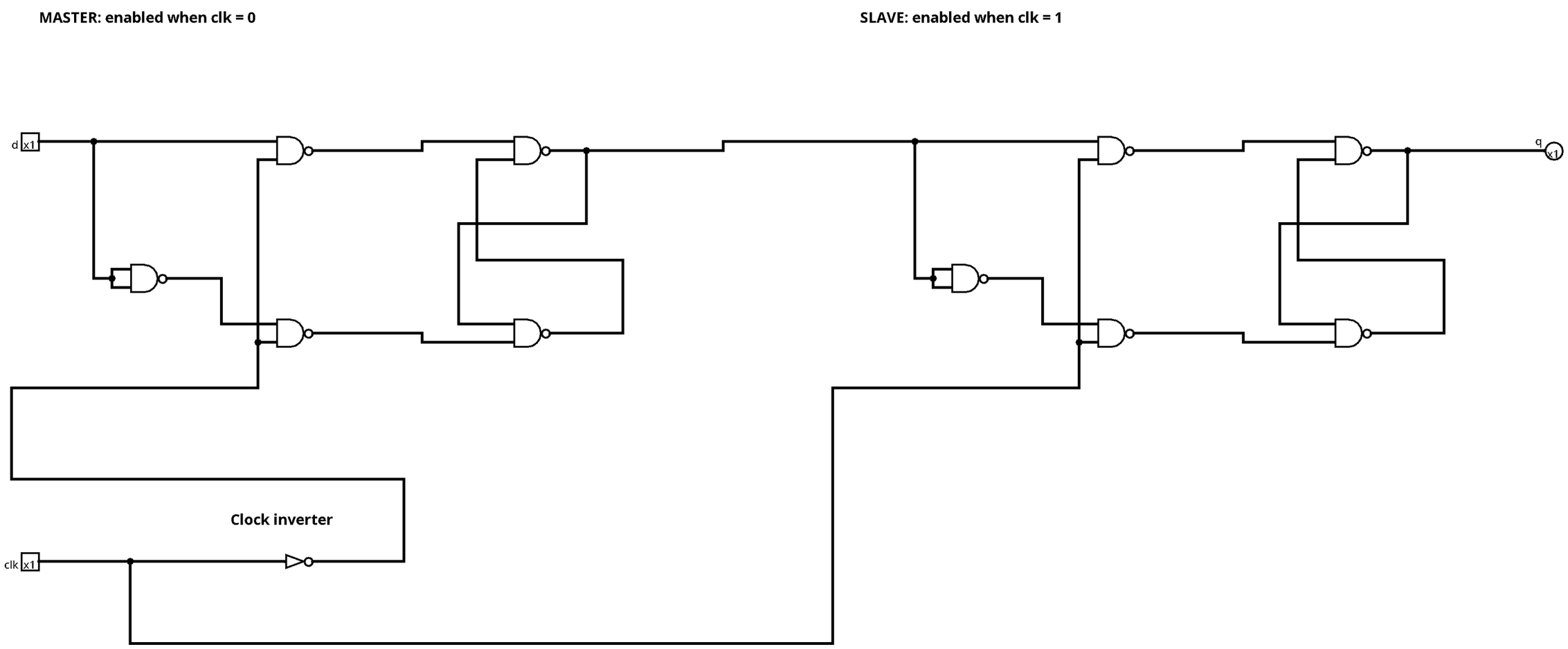
VHDL verification

/workspace/output/assignment3/tb/tb_d_ff_async.vhd:84:3:@170ns:(report note): PASS d_ff_async
simulation stopped @170ns

All RTL and synthesized-netlist assertions passed; generic synthesis completed successfully. Source: vhdl/d_ff_async.vhd. Assertion-based testbench results are shown above.

D Flip-Flop: visible master/slave wiring

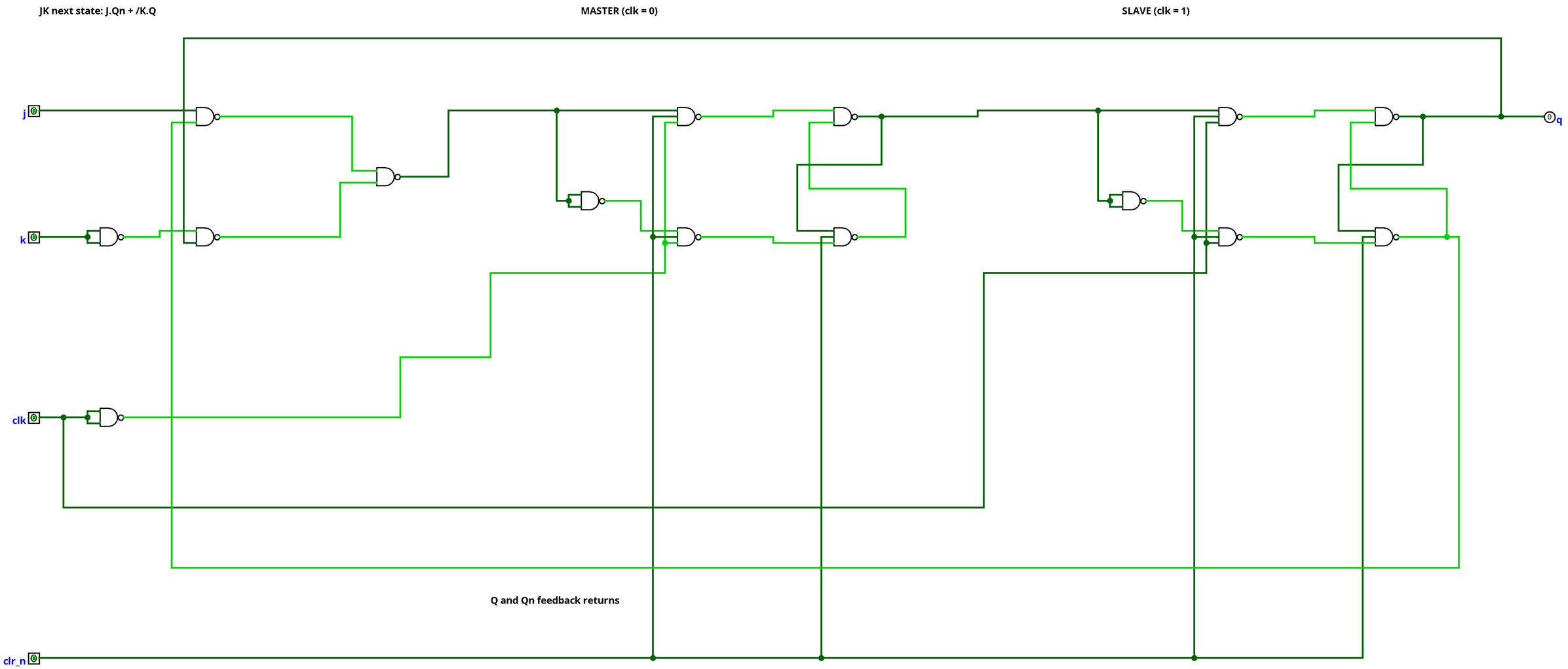
Enlarged circuit drawing | rendered from logisim/d_ff.circ | crossing without a dot = no connection



All five circuits passed the specified checks in Logisim-evolution 5.0.0.

JK Flip-Flop: NAND gates, visible feedback and active-low clr_n

Enlarged circuit drawing | rendered from logisim/jk_ff.circ | crossing without a dot = no connection

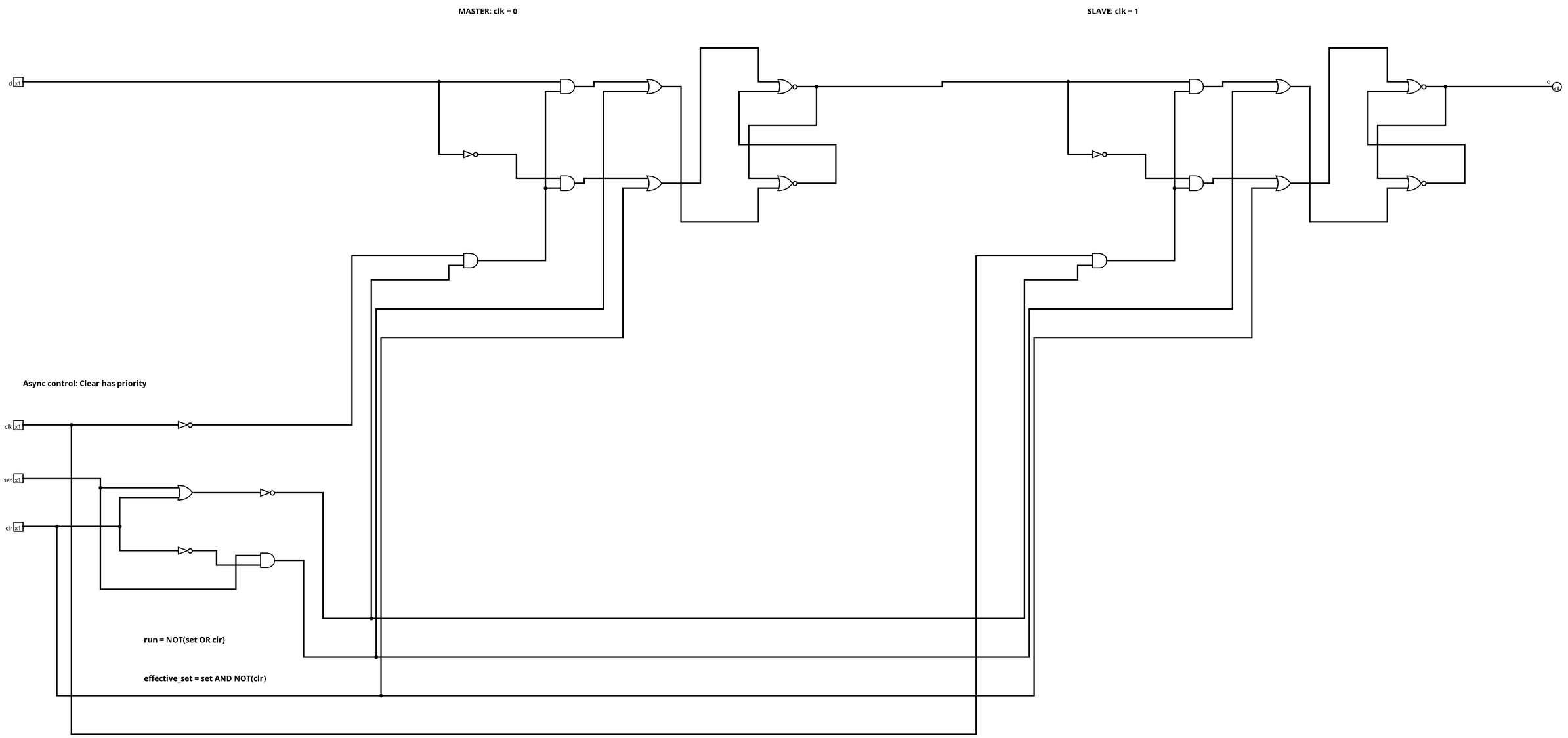


START: clr_n = 0 resets Q. Click clr_n to 1 to RUN.

All five circuits passed the specified checks in Logisim-evolution 5.0.0.

Asynchronous Set/Clear D Flip-Flop: visible control and feedback wiring

Enlarged circuit drawing | rendered from logisim/d_ff_async.circ | crossing without a dot = no connection



All five circuits passed the specified checks in Logisim-evolution 5.0.0.